

B Does the finding of the safety basin generalize to other evaluation metrics and safety datasets?

We expand our experiments to test an additional evaluation metric, LLaMAGuard 2 [26], and another safety dataset, policy-oriented safety evaluation (POSE) benchmark [37]. Our results demonstrate that the LLM safety basins exist regardless of the harmfulness evaluation metrics and safety datasets.

Harmfulness evaluation metrics. We replace the safety keyword detection with LLaMAGuard 2 to evaluate whether the generated output is safe or not. LLaMAGuard 2 is an 8B parameter LLaMA3-based LLM safeguard model. It classifies content as safe or unsafe, and if unsafe, it also lists the content categories violated. As shown in Fig. 5, LLaMAGuard 2 evaluation also shows a basin shape similar to the safety keyword detection.

Safety dataset. POSE benchmark is constructed based on the exhaustive lists of 11 prohibited use cases found in Meta’s LLaMA-2 usage policy and OpenAI’s usage policy. We evaluate the generated outputs using both safety keyword detection and LLaMAGuard 2. Fig. 6 clearly shows that on the new dataset, both evaluation metrics show a similar basin shape.

C Is the capability landscape the same as the safety landscape?

We evaluate on three datasets covering capabilities in math, history, and policy from MMLU [16]. The shape of the LLM capability landscape is drastically different from the one in the LLM safety landscape; these landscapes do not exhibit the same trend, further confirming that the basin shape is indeed unique to the safety of LLM.

We evaluate capabilities using the following three datasets from MMLU: `abstract_algebra`, `high_school_us_history`, and `us_foreign_policy` datasets. Fig. 7 presents the results of perturbing the LLaMA2-7B-chat weights along a 1D-random direction. For controlled comparisons, all datasets are evaluated along the same random direction. We observe that the shape of the capability score varies significantly across different datasets. For example, in the `abstract_algebra` dataset, the model also peaks at $\alpha = 0.2$ (x-axis), while in the `us_foreign_policy` dataset, the model achieves slightly better performance at $\alpha = 0.15$. In contrast, randomly perturbing model weights maintains the safety level of the original aligned model in its local neighborhood, showing a rapid decrease in safety at the brim of the basin. Such drastic changes are not observed in the capability landscape. The gradual changes in the capability landscape align more with the common expectations, but the significantly different shape of the safety landscape is surprising!

D Does the model still generate fluent output when ASR is high?

We conduct additional quantitative and qualitative experiments, which show that LLMs speak fluently even when ASR is high. We measure results quantitatively by using the perplexity on MTBench [52], and qualitatively by listing generated responses sampled along these directions. We evaluate the perplexity of the perturbed LLaMA2-7B-chat model along a random direction using all 80 prompts from MTBench. Fig. 8 demonstrates that the model maintains high fluency (low perplexity), even when ASR is high, except at the extremes ($|\alpha| > 0.4$). Table 3 shows five responses sampled equally along the random direction ($\alpha = -0.5, -0.25, 0, 0.25, 0.5$). At $\alpha = -0.25$, we clearly observe the model speaks fluently but fails to refuse the harmful input.

E The effect of model size on safety basin

We scale up the model size from LLaMA2-7B-chat to LLaMA2-13B-chat. Fig. 8 plots the 1D safety landscape of both models. Interestingly, a larger model size exhibits a wider safety basin, which also aligns with the intuition that a wider basin seems to be more robust and a potential training goal for future LLM training.

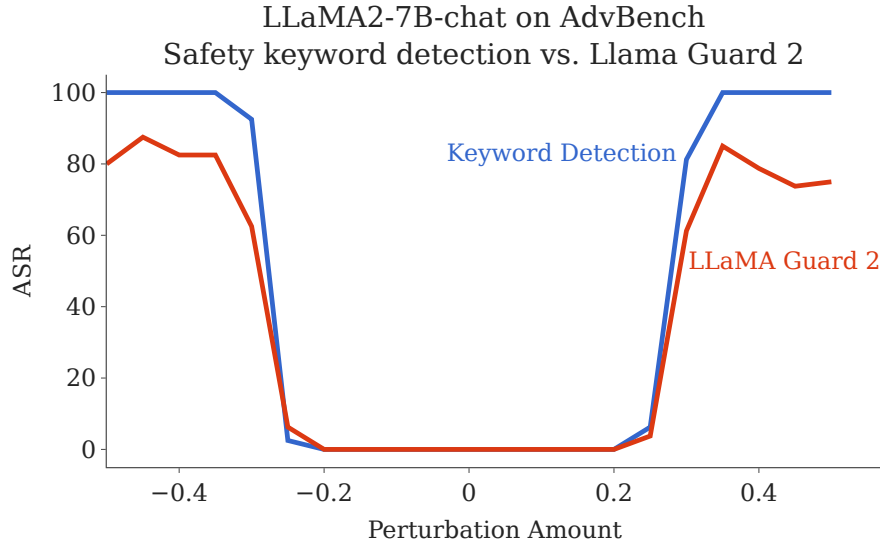


Figure 5: LLaMAGuard 2 evaluation also shows a basin shape similar to the safety keyword detection.

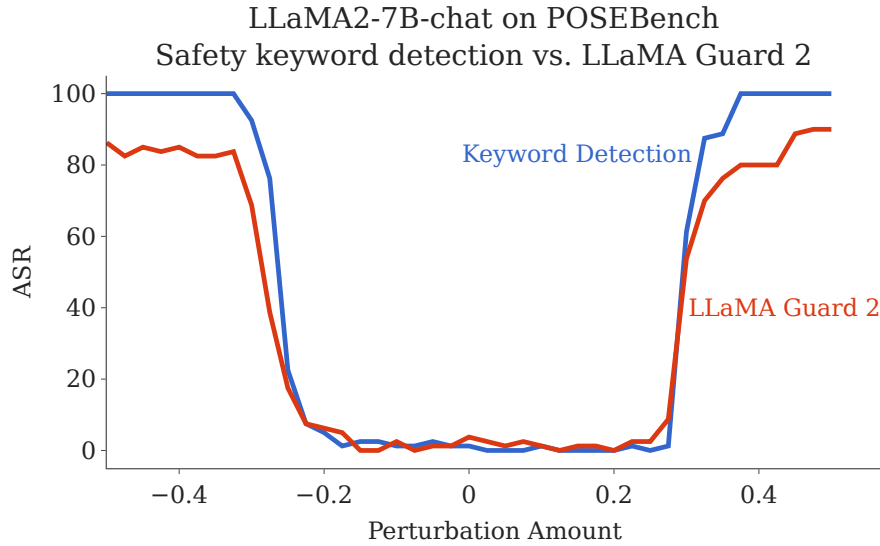


Figure 6: Results on POSE benchmark again verifies the safety basin observed on the AdvBench benchmark. We evaluate the generated outputs using both safety keyword detection and LLaMAGuard 2 and both evaluation metrics show a similar basin shape.

LLaMA2-7B-chat capability landscape on MMLU (5-shot)

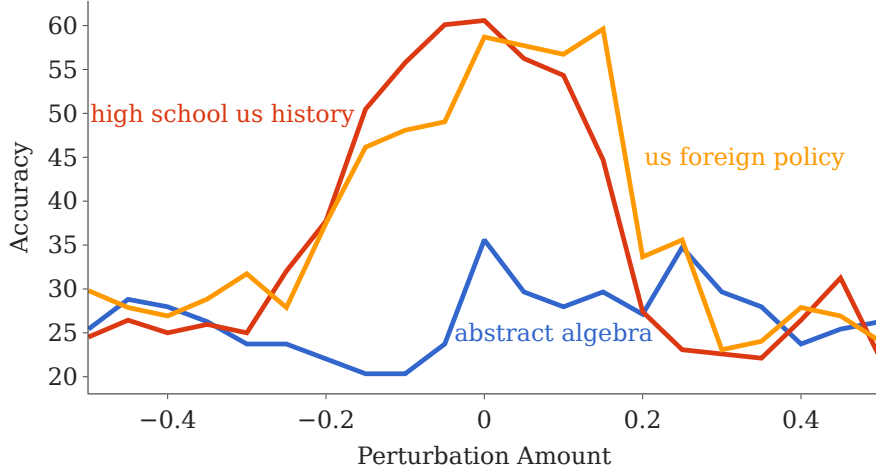


Figure 7: The shape of the capability score varies significantly across different datasets, and differs from the safety landscape. We evaluate capabilities using the following three datasets from MMLU: `abstract_algebra`, `high_school_us_history`, and `us_foreign_policy` datasets, and present the results of perturbing the LLaMA2-7B-chat weights along a 1D-random direction. For controlled comparisons, all datasets are evaluated along the same random direction.

LLaMA2-7B-chat MTBench Perplexity and AdvBench Safety landscape of 7B vs 13B

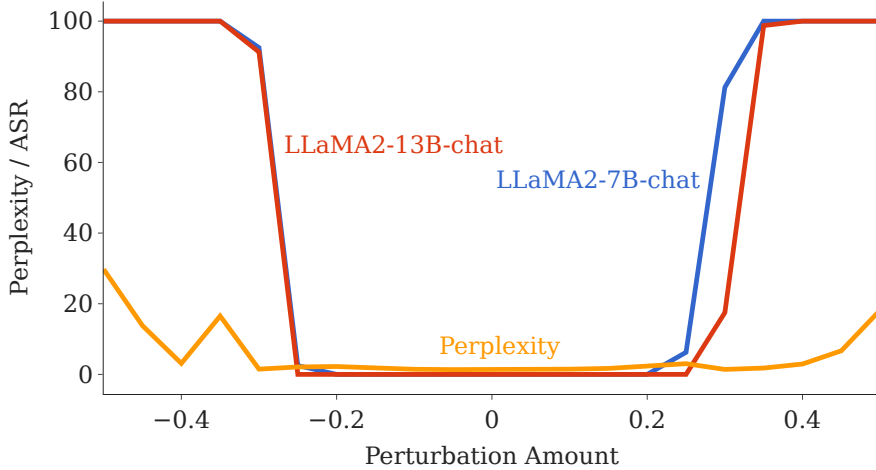


Figure 8: LLMs speak fluently even when ASR is high. We evaluate the perplexity of the perturbed LLaMA2-7B-chat model along a random direction using all 80 prompts from MTBench. We also scale up the model size from LLaMA2-7B-chat to LLaMA2-13B-chat and plot the 1D safety landscape of both models. A larger model size exhibits a wider safety basin, which also aligns with the intuition that a wider basin seems to be more robust and a potential training goal for future LLM training.